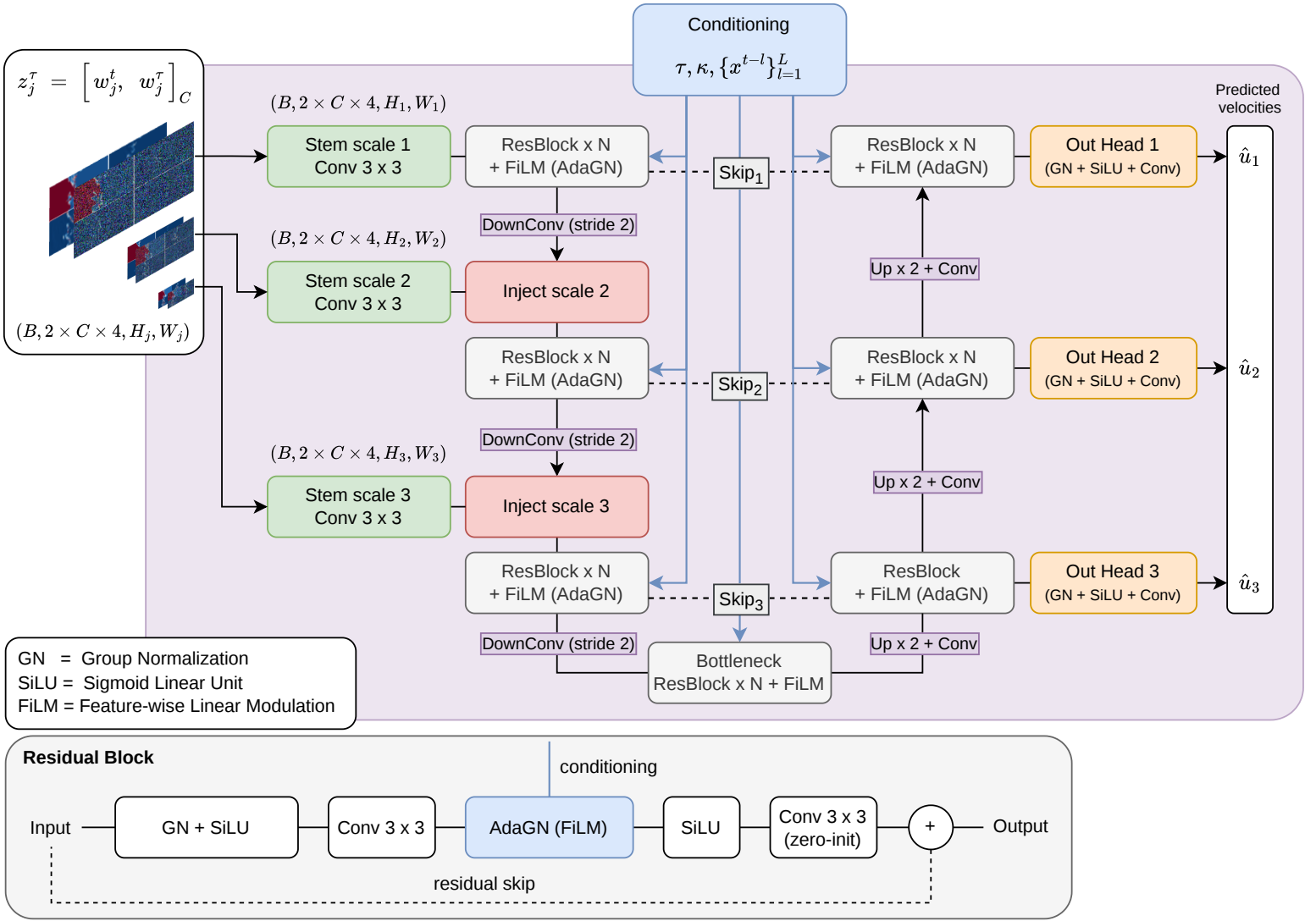

$z_j^\tau = \left[ w_j^t,\ w_j^\tau \right]_C$
$(B, 2 \times C \times 4, H_j, W_j)$
Conditioning
$\tau, \kappa, \{x^{t-l}\}_{l=1}^{L}$
Predicted velocities
$(B, 2 \times C \times 4, H_1, W_1)$
Stem scale 1
Conv 3 x 3
ResBlock x N
+ FiLM (AdaGN)
Skip$_1$
ResBlock x N
+ FiLM (AdaGN)
Out Head 1
(GN + SiLU + Conv)
$\hat{u}_1$
DownConv (stride 2)
Up x 2 + Conv
$(B, 2 \times C \times 4, H_2, W_2)$
Stem scale 2
Conv 3 x 3
Inject scale 2
ResBlock x N
+ FiLM (AdaGN)
Skip$_2$
ResBlock x N
+ FiLM (AdaGN)
Out Head 2
(GN + SiLU + Conv)
$\hat{u}_2$
DownConv (stride 2)
Up x 2 + Conv
$(B, 2 \times C \times 4, H_3, W_3)$
Stem scale 3
Conv 3 x 3
Inject scale 3
ResBlock x N
+ FiLM (AdaGN)
Skip$_3$
ResBlock
+ FiLM (AdaGN)
Out Head 3
(GN + SiLU + Conv)
$\hat{u}_3$
DownConv (stride 2)
Bottleneck
ResBlock x N + FiLM
Up x 2 + Conv
GN = Group Normalization
SiLU = Sigmoid Linear Unit
FiLM = Feature-wise Linear Modulation
Residual Block
conditioning
Input
GN + SiLU
Conv 3 x 3
AdaGN (FiLM)
SiLU
Conv 3 x 3
(zero-init)
+
Output
residual skip